

# Ruthwik Reddy Doodipala

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## SUMMARY

AI Researcher specializing in **Agentic LLM Systems and Foundation Models**. Designs and deploys **end-to-end Generative AI & ML systems** across **HPC** and **AWS** environments. Expertise in **RAG pipelines**, **Autonomous AI Agents**, **PyTorch**, **Distributed Training (DDP)**, and **production-grade model deployment**.

## EDUCATION

**The University of Memphis.**

*Master's in Computer Science, CGPA: 3.98/4*

**Aug 2024 – May 2026**

*Memphis, TN*

## TECHNICAL SKILLS

**Languages:** Python, SQL (PostgreSQL, MySQL), Java

**ML / Deep Learning:** PyTorch, TensorFlow, Scikit-learn, CNNs, GANs, Self-Supervised Learning, Pandas, NumPy

**LLM & GenAI:** LangChain, LlamaIndex, RAG, Hugging Face, OpenAI API, Vector Databases (FAISS, Pinecone)

**Data Visualization:** Matplotlib, Seaborn, Power BI, Tableau

**Deployment & Infra:** Flask, Django, REST APIs, Docker, AWS (EC2, S3, SageMaker), Git, CI/CD

**Systems & Computing:** HPC (SLURM, LSF), Distributed Training (PyTorch DDP), Jupyter Notebooks

## WORK EXPERIENCE

**St. Jude Children's Research Hospital**

*AI Research Intern*

**Jun 2025 – May 2026**

*Memphis, TN*

- Led end-to-end experimentation on **4D fMRI neuroimaging data**, orchestrating **50+ large-scale training runs** on HPC clusters (LSF, SLURM) using **6\*NVIDIA H100 GPUs**.
- Drove model performance improvements by designing targeted ablation studies across architectures and masking strategies, achieving up to **8.26% lift in classification accuracy** and identifying high-impact design choices.
- Engineered an optimized data loading and **caching pipeline** for high-dimensional fMRI data, reducing I/O bottlenecks and improving training runtime by **35%**, enabling faster experiment iteration at scale.

**University of Memphis**

*Graduate Research Assistant, Biomedical Sensors & Systems Lab*

**Jun 2025 – May 2026**

*Memphis, TN*

- Conducted research on **foundation models for neuroimaging**, designing self-supervised and deep learning architectures for representation learning on **fMRI and EEG** data.
- **First author of 2 peer-reviewed research papers** and presented this work at **NeurIPS (San Diego)**.
- Applied **XAI techniques** to improve model interpretability and explain predictions.

**EdgeForce Solutions**

*Data Science Intern*

**Jan 2024 – Apr 2024**

*Hyderabad, India*

- Built end-to-end models for Remaining Useful Life (RUL) prediction on large-scale sensor data.
- Improved performance from **72% to 83%** using feature engineering, hyperparameter tuning, and **5-fold cross-validation**.
- Deployed REST APIs with Flask for real-time inference, reducing manual analysis effort by **40%**.

## PUBLICATIONS

**Region-Aware Reconstruction Strategy for Pre-training fMRI Foundation Models**, *NeurIPS 2025*

[Paper Link](#)

Developed a self-supervised foundation model using masked autoencoder, improving **ADHD classification accuracy by 4.23%**.

**Rhamba:Region-Aware Hybrid Attention-Mamba Framework for Self-Supervised Learning in rs-fMRI** [Paper Link](#)

Developed a hybrid Attention-Mamba model for neuroimaging-based schizophrenia detection, achieving **71% accuracy** with up to **10.66% accuracy** and **6.14% AUROC improvement**, with interpretability using the **XAI** technique Integrated Gradients.

## PROJECTS

**AutoExperiment: Autonomous ML Debugging and Optimization Agent**

[Github Link](#)

*Python, PyTorch, MLflow, LangGraph, Optuna, Docker*

- Built an agentic AI system that autonomously monitors training runs, diagnoses failures (overfitting, NaNs, instability), and iteratively improves models through closed-loop feedback.
- Implemented a multi-agent pipeline (observer, planner, executor, evaluator) for automated hyperparameter tuning and experiment management using MLflow and Optuna.
- Achieved up to **6.8% improvement in model accuracy**, reduced failed training runs by **45%**, and decreased manual debugging time by **60%** through self-optimizing workflows.

**GitScribe: AI-Powered Documentation Agent**

[Github Link](#)

*Python, LLMs, GitHub API, FastAPI*

- Built an AI agent using LLMs and the GitHub API to automatically generate and maintain documentation for repositories.
- Updated documentation within **40 seconds** of a pull request by triggering automated, PR-ready changes via GitHub Actions.
- Reduced manual documentation effort by **~60%** through end-to-end automation of documentation workflows.